

Program 3: Working with Selenium: Maven Project To Connect To Web Servers Using Selenium Dependency

1.Sauce.demo webpage

Step 1: create a maven project on terminal by typing the command **mvn**

archetype:generate -DgroupId=com.example -

DartifactId=MyMavenSeleniumApp01 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

```
samiksha@samiksha:~/report$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp01 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO]
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO]
[INFO] Parameter: basedir, Value: /home/samiksha/report
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenSeleniumApp01
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/samiksha/report/MyMavenSeleniumApp01
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 4.456 s
[INFO] Finished at: 2026-03-17T20:40:07+05:30
[INFO]
[INFO]
```

Step 2: Before we connect to web servers we need to have chrome driver installed. To download chrome driver

To check the chrome version type the command **google-chrome --version** in the terminal

```
samiksha@samiksha:~/report$ google-chrome --version
Google Chrome 145.0.7632.116
samiksha@samiksha:~/report$
```

extract the version and store it in a variable **CHROME_VERSION**

```
samiksha@samiksha:~/report$ CHROME_VERSION=$(google-chrome --version | awk '{print $3}' | cut -d'.' -f1-3)
samiksha@samiksha:~/report$
```

extracts and transforms JSON data

```
LATEST_DRIVER=$(curl -s "https://googlechromelabs.github.io/chrome-for-testing/latest-patch-versions-per-build.json" | jq -r ".builds[\"$CHROME_VERSION\"].version")
```

do install curl using the command

sudo apt install curl

wget https://storage.googleapis.com/chrome-for-testing-

public/\$LATEST_DRIVER/linux64/chromedriver-linux64.zip

Extract and move it to **/usr/local/bin/**:

unzip chromedriver-linux64.zip

```
sudo mv chromedriver-linux64/chromedriver /usr/local/bin/chromedriver
```

```
sudo chmod +x /usr/local/bin/chromedriver
```

Check if ChromeDriver is installed correctly:

```
santiksha@santiksha:~/report$ chromedriver --version
ChromeDriver 145.0.7632.117 (de4e5d6e13b995f5e04137970c0e8ec456017467-refs/branch-heads/7632@{#3101})
santiksha@santiksha:~/report$
```

Step 3 : edit the pom.xml file and add the dependencies

The pom.xml has the code

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/maven-v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>MyMavenSeleniumApp01</artifactId>
  <packaging>jar</packaging>
  <version>1.0-SNAPSHOT</version>
  <name>MyMavenSeleniumApp01</name>
  <url>http://maven.apache.org</url>
  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>
    <!-- https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->
    <dependency>
      <groupId>org.seleniumhq.selenium</groupId>
      <artifactId>selenium-java</artifactId>
      <version>4.29.0</version>
    </dependency>
  </dependencies>
  <!-- Build: Configuring plugins and build settings -->
  <build>
    <plugins>
      <!-- Example: Maven Compiler Plugin to compile Java code -->
      <plugin>
        <groupId>org.apache.maven.plugins</groupId>
        <artifactId>maven-compiler-plugin</artifactId>
        <version>3.8.1</version>
        <configuration>
          <source>1.7</source>
          <target>1.7</target>
```

```

</configuration>
</plugin>
<!-- Example: Maven Surefire Plugin to run tests -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>2.22.2</version>
</plugin>
<plugin>
    <groupId>org.apache.maven.plugins</groupId>

    <artifactId>maven-jar-plugin</artifactId>
    <version>3.0.1</version>
    <configuration>
        <archive>
            <manifestEntries>
                <Main-Class>com.example.App</Main-Class>
            </manifestEntries>
        </archive>
    </configuration>

</plugin>

</plugins>
</build>
</project>

```

Step 4: Edit the App.java

```
package com.example;
```

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
```

```
/**
 * Hello world!
 *
 */
public class App
{
    public static void main( String[] args )
    {
        WebDriver driver=new ChromeDriver();
        driver.get("https://www.saucedemo.com/");
        driver.manage().window().maximize();
    }
}
```

```

    driver.findElement(By.id("user-name")).sendKeys("standard_user");
    driver.findElement(By.id("password")).sendKeys("secret_sauce");
    driver.findElement(By.id("login-button")).click();
}
}

```

Step 5: build and run the maven application

Type **mvn clean install** in the terminal

```

samiksha@samiksha:~/report/MyMavenSeleniumApp01$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MyMavenSeleniumApp01 >-----
[INFO] Building MyMavenSeleniumApp01 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenSeleniumApp01 ---
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenSeleniumApp01 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/samiksha/report/MyMavenSeleniumApp01/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenSeleniumApp01 ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/samiksha/report/MyMavenSeleniumApp01/target/classes
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO]
[INFO] Total time:  11.900 s
[INFO] Finished at: 2026-03-17T21:18:01+05:30
[INFO]
[INFO] -----

```

To execute type the command **mvn exec:java -Dexec.mainClass="com.example.App"**
it opens saucedemo.com and we can login using the credentials

